

Wilf Bates

Scientific Computing & Data Science

Dedicated and enthusiastic MSc student in Scientific Computing & Data Science at the University of Bristol, with a First-Class BSc in Natural Sciences (77%) from the University of Exeter. Skilled and passionate in applying numerical methods and machine learning to life sciences, with a strong drive to translate academic expertise into innovative, real-world solutions.

Education

MSc SCIENTIFIC COMPUTING & DATA SCIENCE • UNIVERSITY OF BRISTOL • SEP 2025 - SEP 2026

- Currently developing advanced skills in coding, machine learning, and high-performance computing, with a focus on applying computational methods to complex scientific problems across disciplines. Training includes modern programming, data science, AI systems and software engineering principles, reinforced through hands-on projects and interactive workshops.

BSc NATURAL SCIENCES • UNIVERSITY OF EXETER • SEP 2022 - JUN 2025

Completed a broad degree spanning Biology, Chemistry, Physics, Mathematics, and Computing, before specialising in interdisciplinary Scientific Computing and quantitative methods. Graduated with a First Class Honours (78%), **awarded the title of Outstanding BSc Graduate** (top of cohort), and received **two Dean's Commendations for Exceptional Performance** (top 5%).

Key achievements and modules include:

- Mathematical & Computational Skills: Scientific Programming in C (82%), Numerics & Simulations (88%), Artificial Intelligence & Applications (77%), (masters) Mathematical Modelling in Biology and Medicine (70%).
- Applied Science: Physics of Climate Change (87%), Physical Chemistry (71%), Energy Materials & Sustainability (86%)
- Advanced Project Work: Led a final-year group project on neural network image analysis, focusing on dataset complexity and CNN hyperparameter optimisation. Took on both organisational and technical leadership while mentoring less experienced team members.
- Undergraduate Research Project (76%) – Investigated computational methods for modelling how stem cells develop into specialised cells using single-cell data. Demonstrated strengths and limitations of current AI tools for mapping cell development.

A LEVELS, GCSE'S • THE LEYS SCHOOL, CAMBRIDGE • SEP 2017 - JUN 2021

- A levels: Maths, Physics, Computer Science (AAA).
- GCSE: 11 GCSE's with grade 9 for English Language & Maths

Employment History

Account Intern • LOW Europe • Jun 2025 – Aug 2025

- Supported wide range of projects: energy, planning, SMEs, and digital competition.
- Developed and presented analyses to senior management, outlining AI solution benefits, feasibility, timelines, and projected costs.
- Designed workflow automation using ChatGPT API and Airtable, improving digital outreach and content management.
- Assisted in exploration of multi-agent system (MAS) automated tender processing EU Commission tenders with NLP.

Head Counsellor • Camp Beckett • JUN 2023 - AUG 2023

- I led a cabin of 8 boys (age 13-15) and 2 staff members, where I was responsible for planning activities twice a day and running classes.
- This required enormous energy and creativity
- Provided invaluable lessons in leadership and team building.

Contact

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<https://wilf-b.github.io/>

Skills

Technical Skills

- 8+ years Python experience
- Strong Mathematical ability
- Applied experience of C, MATLAB, R
- Experience with TensorFlow and other ML libraries
- Data Analytics
- Git, Command line, super computing

Soft Skills

- Communication
- Teamwork
- Enthusiasm
- Independence
- Presenting
- Adaptable

Achievements

- Vice President, Natural Sciences Society: Organised popular events including wellbeing socials, park runs, and extracurricular lectures.
- Volunteer, Raleigh International (Nepal, 10 weeks) – Contributed to building infrastructure and delivering awareness sessions in rural communities, developing strong teamwork skills through challenging and rewarding experiences.
- Charity Running – Raised alone over £2,000 for various charities through long-distance events.
- Head of House & Committee Member (School) – Coordinated external speakers for audiences up to 200; introduced talks and led Q&A sessions.
- Across these roles I strengthened leadership, teamwork, communication, and organisational skills.